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Architectural ecographies

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ABSTRACT: The paper proposes to describe buildings as patches of landscape, drawing inspiration from authors such as A. Tsing and S. Cubitt. While the notions of patch, assemblage, and coordination, rooted in landscape ecology, provide a school of thought to understand buildings as "multi-sited, multi-scalar, and more-than-human", anecdotalism, with its interest in addressing how high-resolution events could promote the comprehension of larger-scale issues, facilitates the tracing of the multiple storylines that intersect in architecture. In response, representational questions arise through processes of entangled maps, here addressed as ecographies, to propose novel practices of registration necessary to understand the entanglement of our discipline with the current climate crisis. To discuss the operativity of these concepts and tools, Ecolonia, a housing development in Alphen aan den Rijn (NL), is taken as an example to test how mapping through "patchy epistemics" and "anecdotal evidence" could radically transform how we describe buildings.

1 REVEALING ARCHITECTURAL DEPENDENCIES

The disruption caused by climate change is shaking architecture at its core, affecting its practice and pedagogy. The profound shift necessary to avert environmental collapse should be accompanied by a novel understanding of architecture, renewing attention to what the modern project has excluded in pursuing its divided model of knowledge (Bauman 1991).

Yet, architectural main production, namely *buildings*, are still fundamentally described and designed to favor simplicity and abstraction over engagement with their terrestrial relations (Moe 2021) and the heterogeneity of our living world. But buildings are entangled with several human and nonhuman actors, with multiple resources and processes from places of extraction, maintenance, end of life, and the political, economic and racial context in which they are produced. Those relations and their ripple effects are hard to grasp, as they happen across heterogeneous scales and time. Understanding, managing and communicating the interdependencies of our built environment is key for recognizing architecture's implications with the so-called Anthropocene and pursuing an ecological agenda.

This paper explores concepts and tools for understanding architectures as entangled accounts of more-than-human actors and processes. Without operating an analytical distinction between nature and society (Moe 2021) or flattening their ontological differences (Bennet 2010), it builds upon the work of a selection of environmental humanities and media studies scholars, such as Anna Tsing and Sean Cubitt, who introduced fertile concepts useful to manage the polyphony of an ecological inquiry. Their approach to unpacking things is tangible, grounded in landscapes and stories and, therefore extremely operative for architectural discourse, as easier to handle in its concreteness, because "thinking with landscapes opens analysis to a constrained multiplicity: the material expands to include the relationships that make places and niches/but not any relation is included" (Tsing 2019).

Considering buildings as landscapes could be a simple yet compelling shift to introduce a framework of complexity and dependency. To do so, we must first unpack a few foundational concepts, rooted in landscape ecology.

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2 BUILDINGS AS MORE-THAN-HUMAN ASSEMBLAGES

Patches are "discrete areas of habitat differing from their surroundings", which exhibit a notable level of intra-species interaction (Dramstad et al. 1996). "Patches might be as large as a national forest or as small as a tree", showing how they can be recognized across scales. Noticing a patch is just the first step in unpacking its aliveness: to understand its *inhabitants* it's useful to recall the ecological notion of assemblage, which describes all the "beings that happen to be in a patch, whatever their relations with each other and with humans" (Tsing et al. 2024).

Anthropologist Anna Tsing has fruitfully stretched those concepts beyond the disciplinary boundaries of landscape ecology, creating a school of thought in which it was possible to understand spaces where different ecologies, cultures, and politics come together without preconceiving their relationship or anticipating their impact, as the consequences of those encounters might be unexpected or unpredictable. Within this framework, patches are "usefully open-ended" (Tsing et al. 2024), and assemblages, that with the mentioned ontological expansion are declined as *polyphonic*, are also described as "open-ended gatherings" of humans and non-humans (Tsing, 2019).

Together with Elaine Gan, she introduces another fertile concept that helps to understand how an assemblage holds together, the notion of "coordination" (Gan and Tsing 2019). "Coordinations are temporal rhythms across varied practices that together produce a new capacity or emergence", therefore following coordinations between different elements of an assemblage clarifies how they relate with one another, and how these relations happen through specific temporal alignment.

Patches, assemblages and coordinations are compelling analytical tools, because they facilitate the study of entangled matters across multiple scales and times, managing the unruliness of the living world with a grounded approach.

Buildings, as mentioned previously, are much more than abstract artefacts: they are imbricated with multiple terrestrial events, industrial transformation, and numerous social relations, happening across a heterogeneous spacetime while still "architects routinely engage modes of epistemic and disciplinary isolation to describe decidedly non-isolated phenomena like buildings and urbanizations" (Moe 2021). This approach has consequences, as it offsets fundamental processes, keeping them abstract and unknown. As "ecological and social degradation are inherent to a model of knowledge and practices based on divided descriptions of otherwise consistently conjoined terrestrial phenomena" (Moe 2021), if architects aim to address the core social, ecological, and political crisis of our times, these numerous invisible-yet-connected factors must find ways to be integrated into architectural knowledge and education.

Thinking with landscapes could radically transform how we both analyze and narrate *buildings,* promoting an approach that favors a grounded connectedness instead of an abstract autonomy. How would the understanding of *buildings as patches*, made by a heterogeneous assemblage of actors and processes, challenge not only their perceived boundaries but also our modes of description? Analytical methods and representation techniques could facilitate this ontological shift.

3 MOVING ACROSS SCALES, SITES, TIME, AND POINTS OF VIEW: THE OPEN-ENDED QUESTS OF ENTANGLED MAPPING

Considering *buildings as patches* mobilizes new visual and representative quests to follow and describe the heterogeneity of entanglements that mix through architecture.

Architecture drawings, especially technical ones, "make a firm distinction between what is important and what can be ignored, what is the concern of the client, engineer, builder, and what is not." (Material Cultures 2022). London-based practice Material Cultures argues that the typical focus of architectural drawings creates a form of confinement that prevents a deeper engagement with a wider economic, climatic, and ecological understanding of architecture. While this reflection is specifically applied to construction drawings for the very nature of their scope, an abstract and isolated idea of what is architecture gets projected beyond those documents. To refamiliarize ourselves with a wider understanding of what is *truly* a building requires a conceptual step back, venturing on questions of representation and worldbuilding, typically addressed by *maps*.

"Maps don't simply show the world as it is, maps create specific kinds of territories - and specific ways to move through the world" A. Tsing, J. Deger, A. Saxena and F. Zhou argue in their recent publication, Field Guide to the Patchy Anthropocene. Representation, in fact, is never passive or innocent as "it shapes what we attend to, what we avoid, and what we overlook all together". Within Feral Atlas, the authors have developed a series of maps exploring a plurality of perspectives, scales, and representational practices, where the act of drawing is "analytically committed in depicting grounded material relations" critically contributing to contrast mainstream perspectives, going beyond questions of representation (Tsing et al. 2024).

Mirroring the above reflection on the architectural realm, these maps could disrupt and challenge the convictions of modernism, pointing us towards the complexities and frictions of architectural assemblages, and suggesting relevant practices of registration necessary to understand the profound entanglement of our discipline with the current climate crisis. "When we talk about description and representation, we do so with a sense of the world-making force of such knowledge practices" (Tsing et al. 2024), recognizing that these acts are not just passive reflections, but they actively influence values and perceptions.

To allow this transformation, a few operations are needed.

3.1 *Transcalarity, multitemporality and multiplicity*

To unpack the physical and political constitution of a building (Moe 2021) scale-shifting observations are fundamental. Attending to different actors and entities mobilizes different scales of inquiry, as they emerge through the defining features of what is observed. For example, following a material through its extraction processes, or non-human actors through their temporal coordination, requires fundamentally different focuses and attention. The concept of *scale*, in this contest, is not anymore a tool of control and precision but helps to discover and recognize constitutive relationships. By tracing various connections across scales and time, we activate multiple storylines. In these maps, temporal considerations are vital, as *buildings as patches* are the spacetime where a series of events, procedures and processes mix. As "different kinds of times are necessary to understand the dynamics of each patch", it's relevant to consider various approaches to temporal analysis and narrations (Tsing et al. 2024).

Anecdotal methods of inquiry, rooted in the understanding of high-resolution events, provide "depth and color" to question large hypotheses through specific experiences (Cubitt 2013). Anecdotes are operative *knots* to weave and unweave the multiple storylines that intersect in such events. "The insistent materialism of the anecdotal method drags us over and over back to the grit of actuality. [. . .] The anecdotal method makes it impossible to ignore the excluded and the effects of exclusion. It forces us to confront the materiality of people, things, and events, and therefore makes us understand that in any event the human cannot be separated from the technical, physical, or organic environments. In this regard, anecdotalism is an ecological approach and in that sense is anti-humanist" (Cubitt 2013). Anecdotes are great tools for examining the heterogeneity and the more-than-human constituency of a patch and for grounding this inquiry into something extremely specific. This diversity of concerns should be our focus: considering *buildings as patches* that gather a *polyphonic assemblage* of different actors and entities, and approaching them through a*necdotal evidence* allows us to discuss a multiplicity of phenomena without being overwhelmed by their ontological differences. Building materials and their anthropogenic effect, legislative frameworks, ethnographic considerations, non-human subjects, etc., all mix and meet in the specificity of anecdotes.

To truly embrace the empiric epistemologies of anecdotalism, stories need to be concrete and experiences specific, deepened through their mundane anecdotes.

4 ECOLONIA: EVERYTHING IS LANDSCAPE

Ecolonia in Alphen-aan-den-Rijn, in the Netherlands, is a sustainable development of 101 houses built between 1989 and 1993 by the Dutch Environmental Agency, Noven, and a risk-bearing social developer, the Bouwfonds. The masterplan was managed by Lucien Kroll, who

coordinated nine teams of architects responsible for designing different housing proposals, developed with environmental concerns considerably higher than the norm. Those requirements were based on three directives from the National Environmental Policy Plan: energy expansion, integrated life-cycle management and quality promotion, which were further specified in more tangible design strategies (Novem 1995). Kroll's scope was to support and facilitate a process of decision-making that could feel disorderly natural, "to achieve maximum complexity within the limits of circumstances", approaching the development as a landscape in which its inner rules and relations are formed by a loose combination of needs and facts. (Kroll 2012).

Lucien Kroll was a special architect. As he explains in his essay "Tout est paysage" (Kroll 1998), he embraced a *landscape* perspective, as its disciplinary lens facilitated a design attitude attentive to plural agendas and longer durations, stretching the scope of an architecture project well beyond the completion of its construction. As a matter of fact, Kroll was a strong opponent to the abstraction, homogenization and violent forms of exclusions enforced by modernism, especially criticizing its intention to streamline, taylorize and simplify all sorts of relations. Married with Simone Kroll, a landscape architect, he designed projects as *healthy* patches, able to thrive over time and grounded in a basic understanding of ecology as the "science of relationships" (Kroll 1998).

Krolls writes about Ecolonia in different moments, the latest in 2012, 20 years after the project's completion, wondering if the depth of such an experiment is just starting to become more recognized (Kroll 2012). In fact, the project, while pursuing a simple idea, was radically innovative: the intention was to demonstrate how an energy-sensitive and environmentally conscious construction could be achieved within the existing frameworks of Dutch housing regulations.

Noven and Boufunds invested considerable effort in bringing together a team of experts to support Lucien Kroll and the nine selected architects, creating a framework of clear communication and unambiguous feedback with ten different agencies to provide testing support and facilitate a flexible approach to current regulations when needed. Kroll's role was to ensure the mutual coherence and coordination between the housing proposals and their adherence to the environmental directives but, whiting this framework, the nine teams were given relative freedom of decision-making, with only the budget as a hard limit (Noven 1995).

Given the ecological qualities of the project and its architects' declared agenda, Ecolonia seems a relevant example of a *healthy patch* worth a deeper analysis to better understand the nature of the relationships that hold it together. With reference to the notion of "patchy epistemics" (Tsing et al. 2024), we propose an exercise of entangled mapping, tested through the *Ecography* of Ecolonia *(Fig. 1)*.

5 TOOLS UNPACKED: JUXTAPOSING STORIES, MATERIALIZING MOMENTS

Patchy Epistemics are "ways of assembling knowledge that make use of a diversity of ways of observing, recording and participating in Anthropocene worlds", by resting on three core commitments "(1) think in, and from, place; (2) embrace a variety of descriptive modes and (3) cultivate connections across difference"(Tsing et al. 2024).

To practice patchy epistemics, the above-mentioned authors suggest specific methodologies of noticing. "Piling is a knowledge practice of gathering up and holding into varying perspectives" a not-so-organised assembly of materials, "with a critical intent that resists the urge to tidy everything into neat and homogenizing bundles". By *piling up* information of different nature, new critical and analytical traction is gained by simply seeing together aspects that were previously separated. A specific kind of piling is the practice of *juxtaposition*, which literally means putting objects side by side. "As a compositional or curatorial method, juxtaposition activates the gap between things. It creates a side-by-side relational tension in which objects and ideas are both together and apart. [...] New and often unexpected connections become evident while distinctive differences remain in view." (Tsing et al. 2024)

To craft our *Ecography, (Fig. 1)* we apply juxtaposition both as an analytical tool and compositional technique, to study the heterogeneity of entanglements that mix through the case of Ecolonia, celebrating the relevance of this variety of concerns.

3D-modelled anecdotes, weaving together aspects of technical, physical, economic, or organic nature, are placed side by side with more traditionally disciplinary documents, such as technical architectural drawings. We see juxtaposition as a technique to operatively support the understanding of *buildings as patches*, able to congregate the polyphonic assemblage that architecture mobilizes through different scales of inquiry and temporalities, while maintaining a side-by-side relational tension to a relevant, even if abstracted, body of knowledge tradition.

As stories depend on their details, the diverse materiality gathered through anecdotal evidence demands a less selective representational approach, able to record and communicate the complex and messy nature of reality. Highly detailed and textured 3d models, depicting the heterogeneity of such anecdotal assemblages, and site-specific 3d scanning are aggregated through software and hardware to help manage such complexity without abstracting their properties (*fig. 2*).

Tsing's theory, free from an ideal of purity "provides traction with the material under analysis, rather than being imposed on it from without" (Tsing et al. 2024), and so a critical depiction of Ecolonia emerges from the combination and the articulation of its diverse stories.



Figure 1. The Ecography of Ecolonia as a test of entangled mapping rooted in *patchy epistemics,* QR code to Figure 1 in high res (https://imgur.com/XUOgGru). Figure 2. Piling and juxtaposing different bodies of knowledge through detailed 3D scanning and 3D models (images by the authors).

6 RECONSTRUCTING ANECDOTAL ETHNOGRAPHIES

What follows is the informal reconstruction of a series of ethnographic anecdotal accounts to deepen the understanding of Ecolonia as a *patch* that gathers a polyphonic assemblage of different actors. Those stories facilitated the comprehension of the *coordinations* happening between materials, people, things, and events, exemplifying how entangled questions could be unpacked and narrated from specific occurrences. In a similar way, the visual process of mapping started from those anecdotal knots, first through *3D piling* and then through *juxtaposition* with Ecolonia's technical documentation. Recalling Tsing's "field guide approach" for "doing theory," we see the proposed *Ecography* as an initial and tentative example, to test "storytelling practice as tools to notice" buildings as landscape patches (Tsing et al. 2024).

While all the narrated events and information are derived from a series of informal interviews held by the authors in July 2024 and from Ecolonia's original documentation (Noven 1995), here the names of the dwellers are made fictitious for the sake of privacy. To allow the authors a more situated writing exploration, to fully engage the potential of anecdotes in providing "depth and colour" (Cubitt 2013), the three short stories are written recalling the authors' personal experience with the use of direct discourse in quotes, to combine smoothly very different elements, like descriptions, technical arguments, subjective impressions, and more.

So, let's turn to Ecolonia. . .

6.1 *Ecolonia, the first anecdotes*

Wood, clay, cellulose, cork, soil, helophytes, water. What right now could be celebrated as a novel example of regenerative architecture has just turned 30. Ecolonia presents itself as

a built landscape of heterogeneous houses and a few small businesses organized through narrow streets, courtyards, squares, cul-de-sacs, gardens, canals, and a pond.

Tjerk, was one of the nine architects responsible for the development. He currently lives in China but is back in the Netherlands just for the summer. Recalling the early days of his career, the making of Ecolonia felt like a collective experiment that manifested through a series of fortunate coordinations. He is satisfied with his contribution: a series of prefabricated timber frame houses that embraced circularity. "Every one of the nine teams of architects had a focused theme and, in my case, I had to put extra attention to limiting water consumption and utilize biomaterials". Tjerk's timber houses aged generally well: the untreated red cedar planks, imported from Canada, have weathered but they are still the original in almost all houses, while the more locally sourced beech wood frame windows have been replaced. The innovative cellulose insulation, from Germany, made from shredded old newspaper and used in the roof and non-load-bearing facades, was generally successful but it caused a few unexpected complications "we sprayed it on-site inside walls and roof, mixed with water to be more cohesive. It was fine but when it dried it shrank, so we needed to add some rockwool to fill the gap," he recalls "we should have insulated the facade elements in the factory and just assembled on-site".

"No problem with the insulation" confirmed Daan, who moved to Ecolonia 20 years ago, "we didn't do any upgrades to the walls. But we expanded the rear drought portal, as the project provided a ready-to-use plinth" he adds, looking at his winter garden addition, "many residents have done it over the years". Tjerk recalls his intention to provide support for future growth, both for humans and non-humans, and how certain decisions were maybe a bit miscalculated. "The corner claddings, to support creepers, are a bit optimistic. We need careful gardening maintenance to avoid the ivy roots breaking the timber planks" Daan explains, "that's why many neighbors have cut it out, but my wife loves basically any plant." The garden is flourishing with hortensia, lilac, apple trees, wildflowers, and also reed and cattails next to the water edge: "We picked this house because the connection with the canal: there were many frogs at the time, but they are completely gone now, taken over by American lobsters. They are everywhere, also in the big pond. My neighbor catches every week, maybe this is what changed the most since we moved here".

When asked if he could point to any dweller who lived in Ecolonia from the beginning, Daan suggests approaching a neighbor on the other side of the water basin. His name is Finn, an ex-tour operator trying to set up the very first Ecolonia architecture excursion, who, over the years, has been carefully collecting the project's original documentation (*Fig. 3*).

We follow the Ecolonia's assemblage in its historical coordinations. . .

6.2 *Revealing the assemblage through anecdotal encounters*

The booklets are spread on the garden table and, at every page flipped, new memories arise. Like when the budgeting led to a new choice of pumps "Water saving was extremely important, so I wanted to be able to use rainwater to flush toilets and gardens, but the cheap pumps were too noisy, and they got almost immediately turned off. "Tjerk explains to Finn "My original selection, a sturdy German brand, would have worked perfectly".

Finn's interest in starting a tour is based on the dwellers' cohesion "We are all very satisfied with being part of this community, we can remember how bare it was when we moved in so, looking around, you can't help but feel proud". Ecolonia looks like a patchwork of landscapes revealing across scales. Knowing Kroll's body of work, this doesn't come as a surprise. In fact, throughout their careers, Lucien and Simone have been extremely attentive in designing careful details to support more-than-human flourishing, such as thick green roofs, connections between gutters, walls and soil, or biodiversity-dedicated zones, have repeated themselves with care and precision.

In Ecolonia, even just looking at the pavement brings wonder: crafty patchworks of bricks, blocks, pebbles and soil blur the boundaries between roads, walkways, and houses. Pavements stop before windows, or next to doors, allowing residents to customize their houses' edges with micro gardens. Lucas and Johanna, the owners of one of the brick houses from Alberts and Yan Huut architects planted a bush of pink lilac 30 years ago, inviting to their house a diverse community of butterflies and bees. "From the very beginning, it felt the right place for us. We didn't just buy a house but a lifestyle, and that's why we are all so invested in nourishing this place. The

ecological *theme* behind our house required longevity of materials and low maintenance, but, apart from that, the project felt really so special that we couldn't dream any better". Lucas and Johanna's house is inspired by the principles of anthroposophy, a movement founded by Rudolf Steiner that, in architecture, developed a language that rejected strong symmetries and straight lines in favor of more natural geometries. "The house is made of load-bearing lime-sand bricks, and all the rooms have at least a wall that isn't straight. It's such a simple move and yet it changes profoundly the quality of all the spaces. And we had to become creative, as normal furniture doesn't fit". Lucas is especially proud of his carpentry work "To create an additional partition wall on the top floor I needed to consider the timber roof, which geometry is quite complex. In fact, because the shape of the house is irregular, here they couldn't prefabricate the floors, like in the other houses. Building was a labor-intensive process, and it involved some waste, and we appreciate the results." It seems that most dwellers have developed a strong relationship with their homes over the years, through a series of smaller and bigger upgrades which all required ad-hoc customizations. The overall heterogeneity of the place supports these changes to happen, instead of rejecting them. "But if you want to take a look at a really special relationship, you need to visit Maria and Jan. They live in the houses with the green roofs, for us the most beautiful of the neighborhood." (*Fig. 4*). *We follow Ecolonia's assemblage in its lively coordinations...*



Figure 3. Ecolonia, the first encounter: Tjerk's house is a lively patch where different actors, materials, and processes meet and mix. Figure 4. Ecolonia, the second encounter: juxtaposing anecdotal evidence and traditional body of knowledge. Figure 5. Ecolonia, the third encounter: the proposed Ecography allows a multiscalar and multitemporal inquiry (images by the authors).

6.3 *Ecolonia holds through more-than-human coordinations*

The green roofs slope down to almost touch the garden. These houses are from architect Renz Pijnenborgh, a dear friend of Tjerk. He had strong ambitions: while minimizing energy consumption, he prioritized the use of biomaterials, the support of biodiversity and a gentle integration into the landscape. The shell of the house is made from 50mm thick sand-lime blocks, milled with vertical slots to host heating pipes. "To decide where to hang paintings without drilling in the pipes we turn on the heating and 'feel' the wall", Maria and Jan explain. "It's an unusual way to know your house." The most spectacular component is the green roof, finished with clay granules, gravel and sedum planting. "It took some time for the vegetation to hold on. During the first storms, the plants blew off, and the roof didn't retain enough moisture so the rainwater drainage, made of chains, became overloaded, and the water blew against the facade, causing algae to grow. Some of the other neighbors replaced the chains with normal pipes, but we kept them. After some years everything settled, and now it works really well." Taking care of the roof requires also a special routine "As we have some pine trees next to our house, we need to make sure that their seeds don't germinate in our roof, so we climb up and check twice a year. In the beginning, the neighbors were mocking us, suggesting to buy a goat to take care of it, but it turned out to be a very manageable and pleasant task. It's just a different gardening." (*Fig. 5*)

Back in Rotterdam, Tjerk is searching for Ecolonia's original brochures in his sister's house. "There were more publications: one was made to collect feedback, another to compare the cost of all the proposed solutions to what was typically used and available in the market. We were occupied by these dilemmas: how much does an environmentally conscious construction cost?

And, more importantly, what savings can an environmentally conscious construction lead to?" By trying to put a price tag on the environmental impact, comparing its additional investment to the market's 'normal' prices, Ecolonia raised fundamental questions on how to transition from a *monetary economy* to a *monetary ecology* (Novem 1995). Different strategies were used to make the proposal financially feasible. For example, if the chosen products were more expensive, they should have had a longer life, a higher residual value, or provided energy savings. This attitude allowed the houses to use solar boilers instead of central heating as, while four times more expensive, they would allow to save a considerable amount of energy required for heating tap water. Also, the walls of the green-roof houses cost approx. 20% more than a normal concrete wall, but the environmental impact was much less pronounced. Moreover, if within the same price range, choices were always made supporting the lower environmental impact: for example, all the foundations reused 20% of concrete rubble waste.

Flipping the pages of such documents, the complexity and sophistication of the project reveal its strong experimental stake, its careful process of registration, and its refreshing relevance in contemporary discourse. Recalling Lucien Kroll's latest thoughts and inspiring stories of more-than-human coordination, we are just starting to unpack *how Ecolonia holds* (Gan and Tsing 2019). The importance of tracing healthy patches.

Can we understand architecture in its complexity, and narrate its entanglement with multiple human and nonhuman actors, resources and processes, passing through mundane everyday moments? Thinking with *patchy epistemics* facilitates the understanding of buildings as landscape patches, made by a heterogeneous assemblage of actors and processes, and their depictions as "multi-sited, multi-scalar, and more-than-human" (Tsing et al. 2024). While still tentative, this example, by bringing together operative concepts supporting ecological thinking, ethnographic observations, and narrative 3D modeling techniques, provides traction for developing methodologies able to visually understand and describe architectural imbrications. In fact, studying examples of buildings as *healthy* patches, like the case of Ecolonia, through and juxtaposing techniques, provides tools to follow and understand the imbricated relations that mix in architecture, beyond both hierarchical and isolationist tendencies. With the recent interest in regenerative architecture, which aims to design buildings that interact positively with natural systems, such as air, water, soil, and biodiversity, helping to restore ecological balance and enhance social resilience, we argue that the above *Ecography* could provide an operative framework of analytical methods and representational tools to map such case studies and understand how they hold together.

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